Mansi Budamagunta

Data Scientist

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Summary

Data scientist with a research background, applying computational methods to complex real-world problems at the intersection of data and public interest — with experience across several domains, including LLM training pipelines for leading AI labs, urban data analytics, transportation networks, and computational neuroscience

Education

Utrecht University *July 2026 (expected)*
MSc Applied Data Science, GPA (so far): 8.5 *Utrecht, The Netherlands*

- Thesis: Computational Policy Analysis of EU Budgets

Indian Institute of Science Education and Research (IISER) Pune *Sept 2021*
BS-MS Dual Degree, Interdisciplinary Major, GPA: 8.9 *Pune, India*

- STEM degree with coursework spanning mathematics, data science, natural sciences, and humanities
- Thesis: Dynamical Processes on Transportation Networks: Effects on Infectious Disease Spreading

Experience

AI Trainer *Jan 2023 – Mar 2025*
Turing

- Worked as a Python programmer on LLM training pipelines for one of the world's leading AI labs
- Contributed to the design of evaluation environments and guidelines across different LLM capability domains — including code generation, reasoning, tool use, web browsing, and computer usage
- Evaluated LLM outputs, identified model weaknesses, and generated targeted feedback within the RLHF loop to improve frontier models
- Led teams of varying sizes, optimising workflows and establishing quality standards

Research Intern *July 2022 – Jan 2023*
Data & Energy in Buildings & Cities Lab, IIT Bombay *Mumbai, India*

- Clustered Indian urban districts using remote sensing, climate, census, and demographic data
- Collected and collated diverse datasets for 96 districts; performed k-means clustering to identify patterns
- Revealed that geographic proximity doesn't always determine similarity of districts
- Presented this research at BuildSys '22: Proceedings of the 9th ACM International Conference on Systems for Energy-Efficient Buildings, Cities, and Transportation

Research Intern *May 2019 – July 2019*
Theoretical Neuroscience Group, Aix-Marseille Université *Marseille, France*

- Analysed connectome-based whole-brain models via numerical simulations
- Contributed to *The Virtual Brain* software (European Human Brain Project)

Skills

Programming: Python, R, SQL, HTML, CSS, JavaScript

ML, NLP & Networks: PyTorch, Hugging Face Transformers, scikit-learn, NetworkX

Data & Viz: Pandas, NumPy, Matplotlib, Seaborn, Plotly

Tools & Infrastructure: Git, Docker, Quarto, LaTeX

Languages: English, Hindi, Telugu (fluent) · French, Dutch (beginner)

Selected Projects

EU Funding & Technological Sovereignty (MSc Thesis, ongoing)

- Investigating EU research funding alignment with technological sovereignty goals (openness and value alignment) using computational text analysis
- Extracting portal data and extending the `sedia-api-fetchers` library to build a reproducible NLP pipeline
- Developing a public-facing dashboard to visualize funding allocation and participant metadata

Multi-Model vs Single LLM for Code Generation

- Evaluated performance trade-offs between modular multi-model pipelines (reasoner + coder) and single large LLMs for code generation tasks
- Implemented local inference pipelines on CUDA-enabled GPUs using PyTorch and Hugging Face Transformers
- Benchmarked structured reasoning capabilities across specialized models using the LiveCodeBench and APPS datasets

Predicting Type 2 Diabetes Risk with Fairness Auditing

- Built an interpretable logistic regression classifier as a case study in responsible AI development and bias mitigation
- Conducted comprehensive fairness audits across demographic groups using Equal Opportunity and Predictive Rate Parity metrics
- Applied EU AI Act, DEDA, and FRAIA frameworks to propose governance protocols, risk assessments, and accountability structures

Infection Spread on Transportation Networks

- Collected Indian census + rail/air transport data for 446 cities
- Simulated SIR metapopulation model to construct an infectious disease hazard map for India
- Contributed to the resulting publication in Current Science (2021) and created the project website

Publications

Budamagunta M., A. Gaur, A. Verma, A. Anand, C. Deb (2022), Clustering Indian Districts Based on Multidimensional Demographic and Climate Data, *Proceedings of the 9th ACM International Conference on Systems for Energy-Efficient Buildings, Cities, and Transportation (BuildSys '22)*, https://doi.org/10.1145/3563357.3566146 *2022*

O. Sadekar, **Budamagunta M.**, G.J. Sreejith, S. Jain, M.S. Santhanam (2021), An Infectious Diseases Hazard Map for India Based on Mobility and Transportation Networks, *Current Science, 121(9), 1208–1215.*, https://doi.org/10.18520/cs/v121/i9/1208-1215 *2021*

Achievements & Extracurriculars

- Member of the Young Innovators honours programme, Utrecht University (2025)
- Part of a 9-member national student delegation to Vietnam for a cultural and academic exchange with universities and government officials, selected by the Ministry of Youth Affairs & Sports, Government of India (2017)
- Volunteer teacher and programme coordinator at Disha, IISER Pune's social outreach club — taught children in slums and villages, and helped design and run some of the club's programmes (2016–2021)
- INSPIRE Scholarship holder (2016–2021), awarded by DST, Government of India